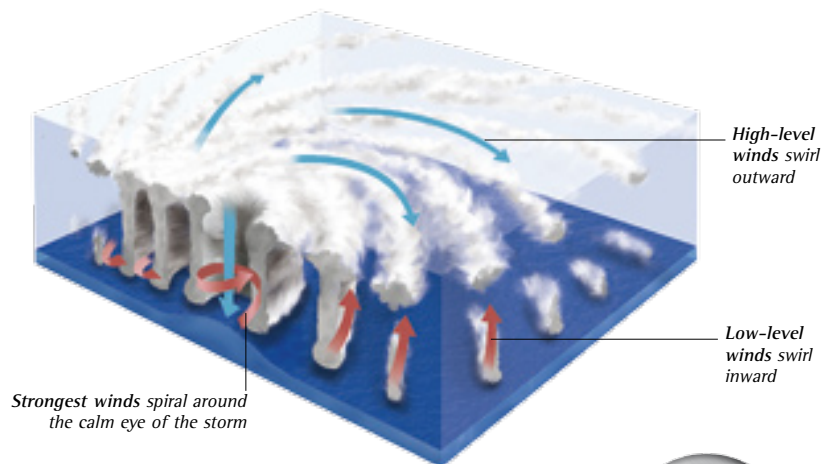


▲ STORM CLOUDS

Clouds form as moist, warm air rises and cools, and the water vapor within condenses into water droplets. This releases heat, warming the air in the cloud and making it rise higher. More vapor condenses, releasing more heat, building huge storm clouds that can be more than 4 miles (6 km) high and cause torrential rain.

▲ WATERSPOUTS

Rising air within a storm cloud causes swirling currents that can turn into a tornado. When this happens over the sea the updraft can suck water up into the cloud, causing a waterspout. Although less violent than a tornado, it can capsize or wreck boats, especially when it collapses and drops its load of water.

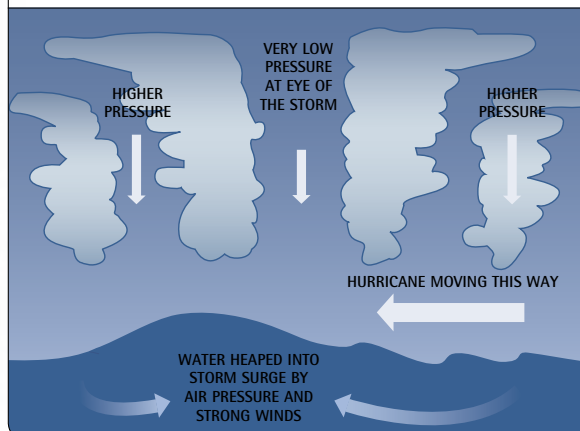


▲ HURRICANES

In tropical oceans where the surface temperature is higher than 81°F (27°C), vast amounts of water can evaporate from the ocean in fast-rising currents that create a zone of very low pressure. This makes the surrounding air swirl into the center of the system at high velocity, creating a revolving mass of huge storm clouds, torrential rain, and extreme winds—a hurricane.

cyclones

STORM SURGE



The atmospheric pressure at the eye of a hurricane is extremely low. This allows the surrounding higher air pressure and converging wind to push ocean water toward the core of the storm. It forms a mound of water called a storm surge, which is pushed ahead of the moving storm centre. If the hurricane approaches land, the storm surge heaps up in the shallow water like a tsunami, creating a wave that may be more than 33 ft (10 m) high. It was this effect that flooded the coastal city of New Orleans in 2005.